3D-printed housing for Indian jawans, built by Indian Army MES in association with Tvasta (Courtesy: Bharat Shakti)

metre (700 sq ft) homes are designed to be disaster-resistant and meet Zone-3 earthquake specifications. They also 3D-printed sanitary blocks with a total built area of about 56 square metres at Jaisalmer.

The startup works closely with the Central Building Research Institute and Structural Engineering Research Centre to ensure that the 3D-printed structures meet required safety standards. But houses are not all that they can build. They have even been asked to build bunkers and parking facilities at border areas, where traditional construction is a challenge due to the harsh weather and hostile conditions!

Similarly, the US Department of Defence (DoD) is building three training barracks in Texas using 3D printing tech, with private partnership. Each barrack will be more than 530 square metre (5700 sq ft) in size, making these the largest 3D-printed structures in the US. It will be printed using a proprietary high-strength concrete, capable of withstanding the harsh desert weather.

Back home in Bengaluru, we hear that an entire three-storied post office is going to come up in Halasuru within a month or so. The project will be completed using 3D printing technology by Larsen & Toubro Construction at just 25% of the normal construction cost.

For *jawans* and green warriors, foods, and *fundas*

3D printing is finding some use or the other in many fields, and it will not be long before these disparate applications unify into large-scale adoption.

Defence. Although 3D printing is not widely used for critical military equipment, it supports the army in many ways. We already read about housing, barracks, and bunkers being built using 3D printing technologies.

3D printing is also used to print armour suits, and parts used in aircraft engines, tanks, and submarines. Recently, the US DoD announced that it plans to pair defaulting suppliers with 3D printing companies that can print metal parts, so they can quickly get over the backlog.

Like in other industries, 3D printing is also used extensively for prototyping and testing military equipment. The Indian army is also collaborating with many Indian 3D printing companies like Chennai based Tvasta Constructions and Mumbai based Divide by Zero.

Researchers at IIT Jodhpur have recently developed a metal 3D printer suitable for repairing and adding additional material to existing components. It can print with metal powders made in India and is considered ideal for printing fully-functional parts for aerospace, defence, automotive, oil, gas, and other industries.

Energy. Benefits like design flexibility, short time-to-market, lighter and better performing parts, reduction in material wastage, elimination of supply-chain issues, and tool-free production are leading several companies in the energy sector to deploy 3D printing solutions. Most market leaders—including BP Global, Chevron, Exxon Mobil, GE Power and Shell Global—are using 3D printing.

Earlier, 3D printing was used only for prototyping and not for end-use components. But, with the ability to print with more materials, it is being used to print various goods like solar panels. According to a report by GE Research, 3D-printed solar panels have proven to be 20% more efficient than traditional ones, and at half the price of manufacturing.

3D printing is also used for manufacturing high-value, high-complexity, low-volume components, like gas turbine nozzles, impellors, pistons, pumps, rotors, and parts for control valves, flow meters, heat exchangers, and pressure gauges. The cost of downtime is very high in the energy sector and failure can occur even in remote areas. The ability to print spares will help reduce downtime without stocking up on spares.

Robotics. 3D printing is a boon for robotics enthusiasts in countries where components are not easily and economically available. Since many component designs are available online under free or open source licenses, enthusiasts with access to 3D printers can easily print these components and use them in their projects. Even if they are unable to own one individually, groups of enthusiasts can pool in and share 3D printers.

In fact, it is this need that motivated the founders of Divide by Zero to float a 3D printing startup. They used to enter a lot of robotics contests when they were in college and were in trouble each time a robot needed a spare. No machine tool shop was ready to manufacture one or two pieces and, even if they did, they used to charge them heavily.

This motivated them to develop an indigenous 3D printer such as the Accucraft i250+. "We thought of making affordable 3D printers for the Indian market and support individuals and organisations to prototype their ideas, play with them and create wonders," said one of the founders in an interview. The inherent benefits make 3D printing attractive to hobbyists and the industry alike.

Openbot by Intel and Atlas by Boston Dynamics are some of the well-known 3D-printed robots, not to forget the emerging range of 3D-printed humanoids, swarms, amphibious and other bio-inspired robots. 3D printing of soft robots from non-toxic, gelatin based material is gaining popularity of late for applications like teaching and training children, food manufacturing,